



RMT — Renewal Management

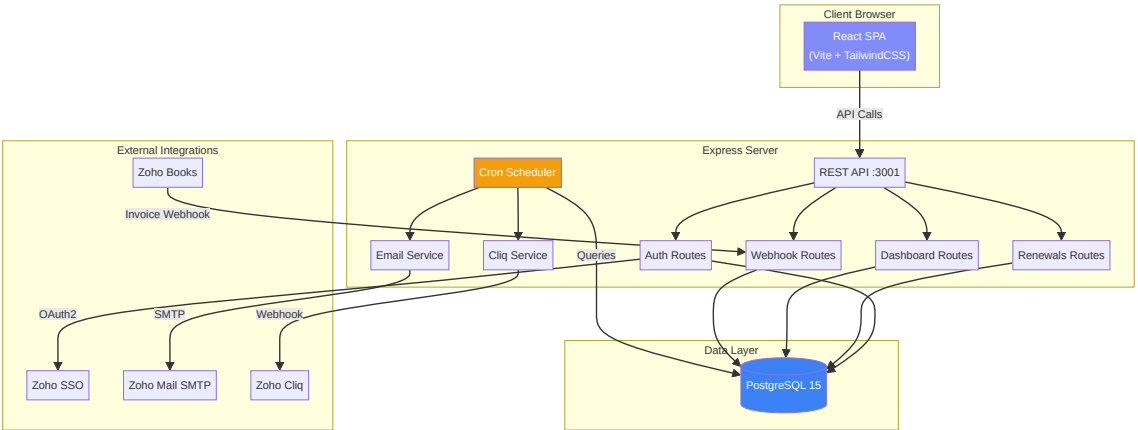
Tool

MarsLab Internal
Generated: June
2026

Complete Project Architecture & Working Flow Documentation

RMT is a full-stack Renewal Management System built for MarsLab to automate the tracking, notification, and lifecycle management of client service renewals.

1. High-Level Architecture

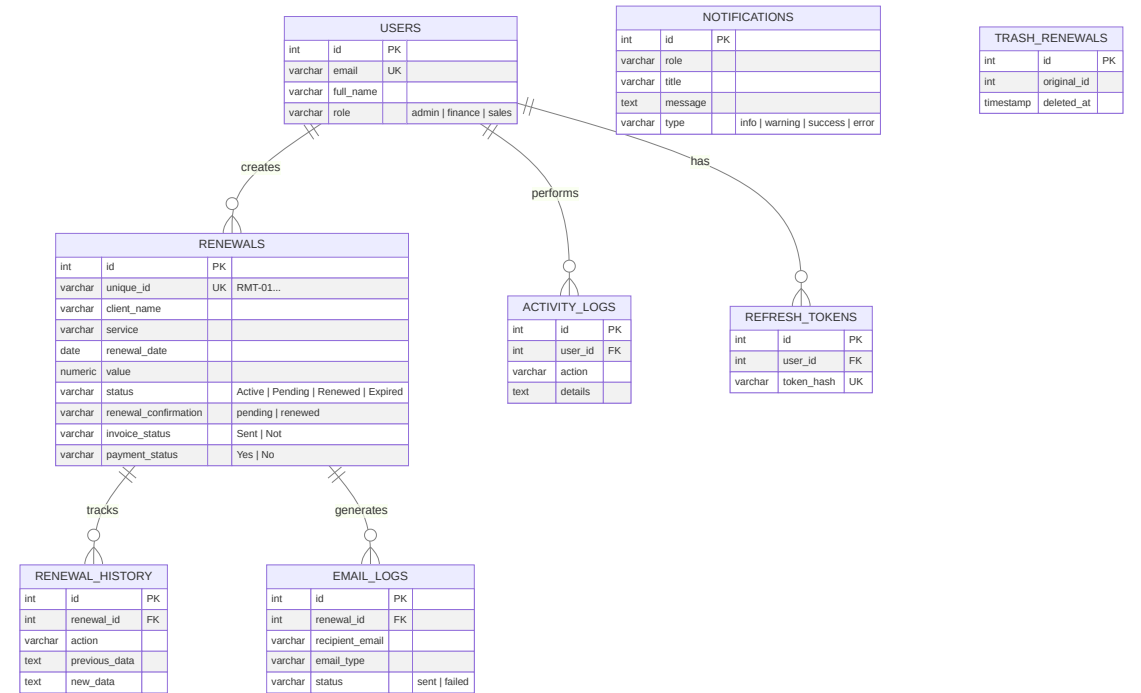


2. Technology Stack

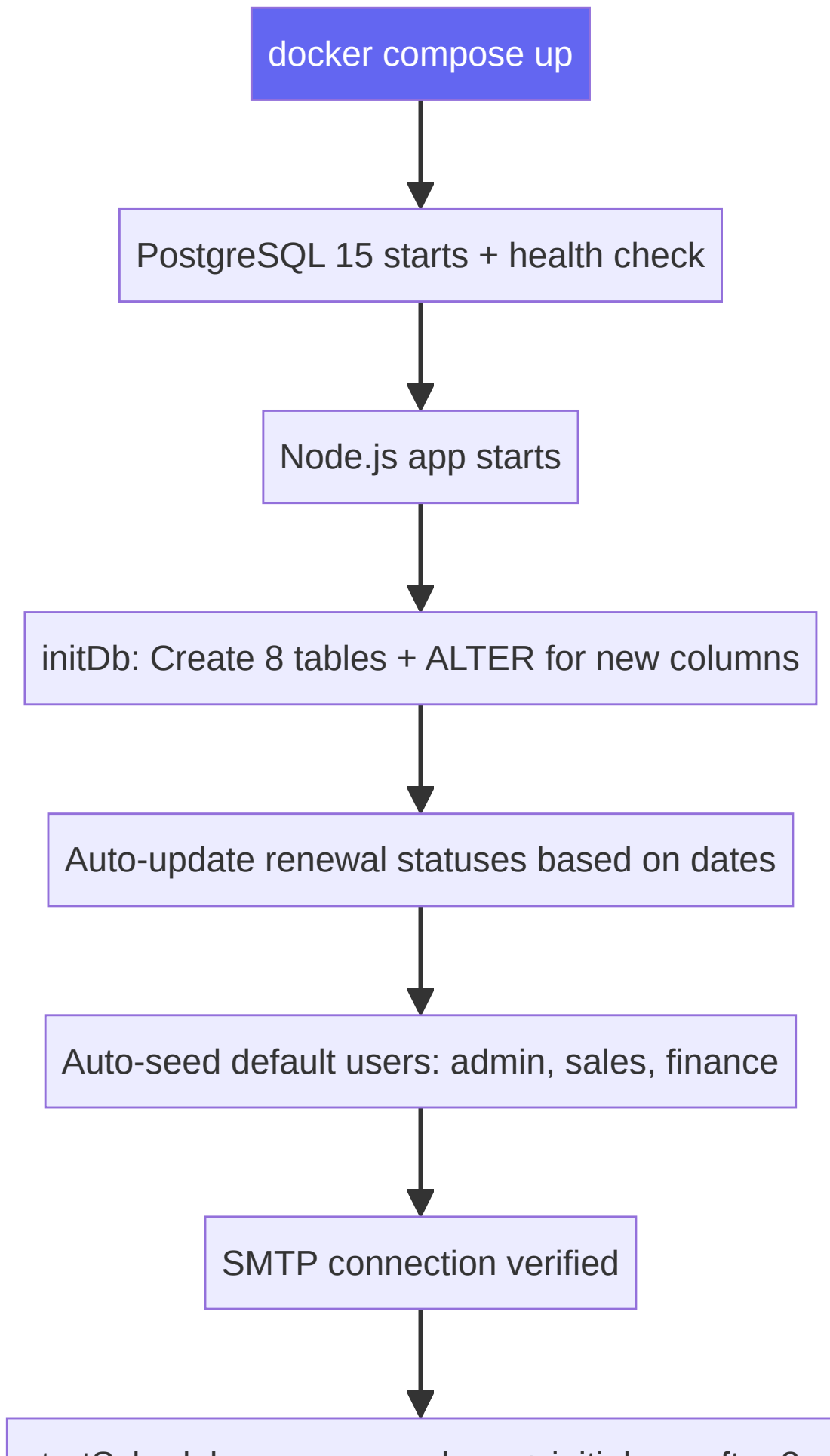
Layer	Technology	Purpose
Frontend	React 18, Vite 5, TailwindCSS 3	SPA with responsive UI
UI Libraries	Framer Motion, Recharts, Lucide React	Animations, charts, icons
Backend	Express 4, Node.js 20	REST API server
Database	PostgreSQL 15 (Alpine)	Persistent data store
Authentication	Zoho OAuth2 SSO + JWT	Secure login with token rotation
Email	Nodemailer → Zoho Mail SMTP	Automated reminders
Notifications	Zoho Cliq Webhooks	Team channel alerts

Scheduling	node-cron	Hourly automated tasks
Invoice Sync	Zoho Books Webhooks	Auto-sync invoice data
Export	jsPDF, xlsx	PDF & Excel reports
Deployment	Docker Compose → AWS ECR	Containerized deployment

3. 🗄️ Database Schema (8 Tables)



4. 🚀 Application Startup Flow

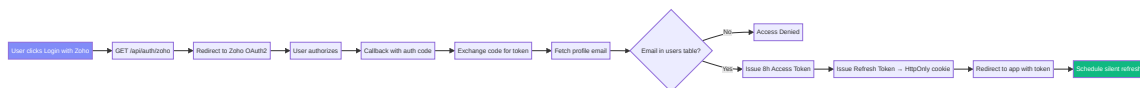


startScheduler: cron every hour + initial run after 35

Server listening on :3001

On every startup, the system auto-recalculates all renewal statuses: **Expired** (past date), **Pending Renewal** (≤ 30 days), **Active** (> 30 days).

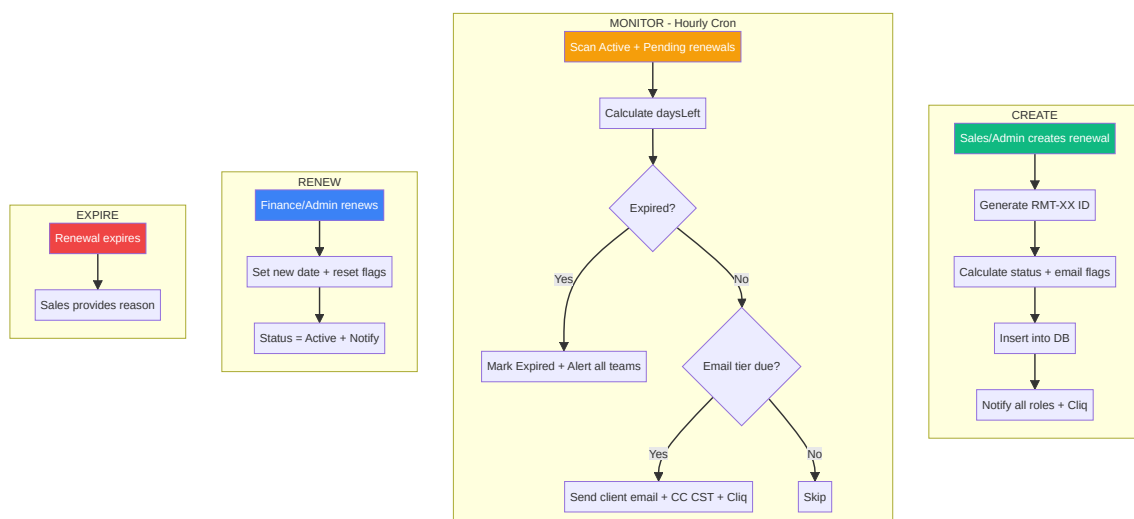
5. 🔑 Authentication Flow (Zoho SSO)



Token Security Design

- **Access Token:** 8-hour lifespan, stored in React state (memory only), sent via `Authorization: Bearer`
- **Refresh Token:** 10-year lifespan, HttpOnly/Secure/SameSite=Strict cookie, SHA-256 hashed in DB
- **Token Rotation:** On each refresh, old token revoked in DB, new one issued
- **Silent Refresh:** Scheduled 2 minutes before access token expiry

6. 📋 Renewal Lifecycle (Core Business Logic)



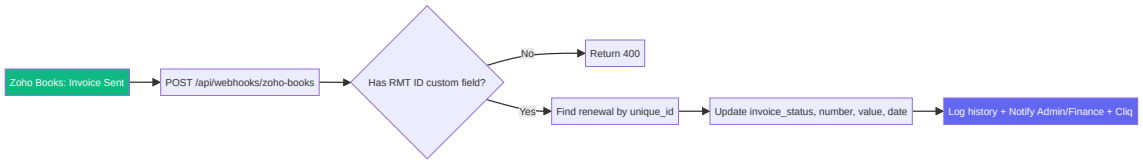
7. Automated Email Reminder Schedule

The scheduler runs **every hour** via `node-cron` and sends tiered reminders:

Days Left	Client Email	Sales Special Email	Cliq Alert
≤ 30 days	✓ 30-day reminder (CC: CST)	—	✓ Both channels
≤ 20 days	✓ 20-day reminder (CC: CST)	—	✓ Both channels
≤ 15 days	✓ 15-day reminder (CC: CST)	✓ "Meet client" to CST	✓ Both channels
≤ 10 days	✓ 10-day reminder (CC: CST)	—	✓ Both channels
≤ 5 days	✓ 5-day reminder (CC: CST)	✓ "Meet client" to CST	✓ Both channels
≤ 3 days	✓ 3-day reminder (CC: CST)	—	✓ Both channels
< 0 (Expired)	—	—	✓ Expiry alert + Email to Admin (CC: CST)

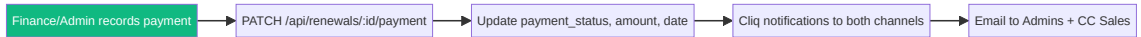
Catch-Up Mechanism: If the server was down and missed a tier, the next run fires the email for any tier not yet sent. It also marks all larger tiers as sent to prevent duplicates.

8. Zoho Books Invoice Sync



Zoho Books invoices must include a custom field named "RMT ID" (e.g., RMT-01) to be linked to the correct renewal record.

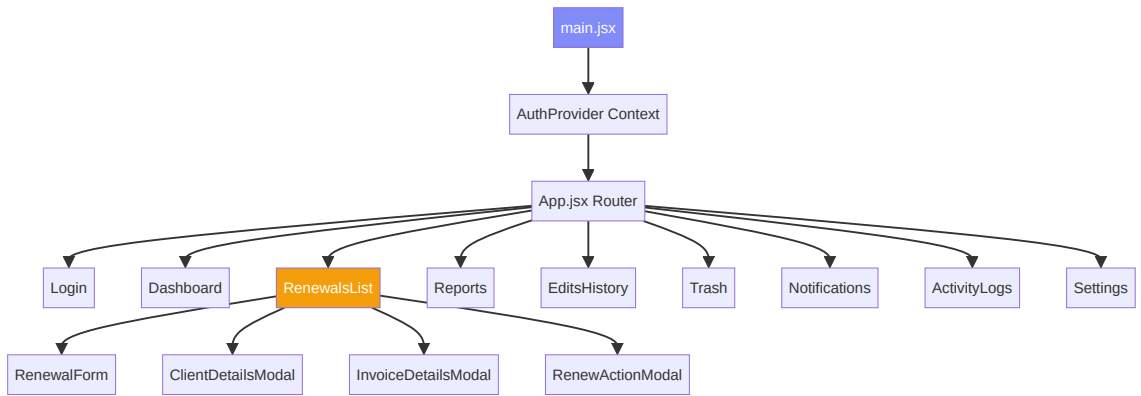
9. 💰 Payment Tracking Flow



10. 🔒 Role-Based Access Control (RBAC)

Action	Admin	Finance	Sales (CST)
Create renewal	✓	✗	✓
Edit renewal	✓	✓	✓
Renew client	✓	✓	✗
Delete → Trash	✓	✗	✗
Restore from Trash	✓	✗	✗
Permanent Delete	✓	✗	✗
CSV Bulk Import	✓	✗	✗
Update Invoice	✓	✓	✗
Record Payment	✓	✓	✗
Follow-up Updates	✓	✗	✓
Trigger Scheduler	✓	✗	✗
View Trash	✓	✗	✗

11. 🖥️ Frontend Architecture



Pages Overview

Page	Purpose	Access
Dashboard	KPI cards + status/monthly charts	All roles
RenewalsList	Main data grid with search, filters, pagination, actions	All roles
Reports	PDF & Excel export of renewal data	All roles
EditsHistory	Audit trail of all record modifications	All roles
Notifications	In-app notification center (role-based)	All roles
ActivityLogs	System-wide activity audit log	All roles
Trash	Soft-deleted records: restore or permanent delete	Admin only
Settings	Password change and profile	All roles

12. 🚢 Deployment Pipeline



Docker Compose Services

- **app** — Node.js server serving both API and static frontend on port 3001
- **db** — PostgreSQL 15 Alpine with health checks, persistent volume, and init script

13. 🔑 Key Configuration (.env)

Variable	Purpose
ZOHO_CLIENT_ID / ZOHO_CLIENT_SECRET	Zoho OAuth2 SSO credentials
ZOHO_REDIRECT_URI	OAuth callback URL
JWT_SECRET / REFRESH_SECRET	Token signing secrets
PG* vars	PostgreSQL connection config
EMAIL_ENABLED	Toggle email sending (true/false)
SMTP_HOST / PORT / USER / PASS	Zoho Mail SMTP config
ZOHO_CLIQ_WEBHOOK_URL	Finance Cliq channel webhook
ZOHO_CLIQ_SALES_WEBHOOK_URL	Sales Cliq channel webhook

14. 💡 Key Design Decisions

🔒 Zoho SSO (No Passwords)

Users authenticate via Zoho org account. The users table is a pre-seeded allowlist.

🍪 HttpOnly Refresh Cookies

Refresh tokens stored as HttpOnly cookies — never accessible to JS, eliminating XSS theft.

🕒 Hourly Cron with Catch-Up

Uses $\leq$ comparison with 'No' flags. If server was down, next run catches up without duplicates.

🗑️ Soft Delete → Trash

Renewals move to trash_renewals. Only admins can restore or permanently delete.

📄 Zoho Books Webhook Bridge

RMT ID custom field in Zoho Books invoices creates a seamless accounting-to-RMT bridge.

📡 Dual Cliq Channels

Every event posted to both Finance (rmt) and Sales (rmtsales) channels.

